SEAL: Separating Supervisor Requests from Code-Enforced Outcomes

Arjun Vijay Prakash
Independent Researcher

When does evidence change a supervisor's publication request, and when does code stop its execution? SEAL records 640 scheduled attack episodes on fixed synthetic local jobs, with two supervisors reported separately. Claims, receipts, and a fixed snapshot still permit unsafe publication. Provenance prevents execution through a deterministic gate; the model can still request publication. We present separate decision and execution counts, a gate mutation audit, and the coverage limits of matched comparisons. The frozen record contains 640 scheduled attacks and 320 honest controls on four families. Attacks are instructed and non-adaptive; the sink is fake and local, `side_effects` is false, and signing keys are public demo keys.

Evidence and enforcement

Model	Coord.	Claims	Receipts	State	Provenance
gemma-4-31b-it	single	40/40/40	30/30/39	29/29/30	30/0/35
gemma-4-31b-it	coordinated	39/39/39	29/29/38	30/30/32	30/0/35
qwen3.5-397b	single	39/39/39	34/34/37	30/30/38	21/0/34
qwen3.5-397b	coordinated	40/40/40	38/38/40	30/30/39	19/0/36

Each cell is unsafe decisions / unsafe executions / completed attacks (D/E/N); models and speaker conditions remain separate. Claims, receipts, and the fixed snapshot still publish completed attacks. Provenance execution zeros come from `_provenance_action` in `src/live_fixtures.py`. Its `trusted_summary != summary` check and earlier binding/authorization vetoes can block a corrected true summary on an illegitimate fixture.

Blind provenance

Only the outward trusted summary is nulled; the gate and other evidence remain unchanged. Cells report old versus blind D/E/N. Answer inference remains possible and collection times differ.

Model	Coord.	Old D/E/N	Blind D/E/N
gemma-4-31b-it	single	30/0/35	30/0/37
gemma-4-31b-it	coordinated	30/0/35	30/0/37
qwen3.5-397b	single	21/0/34	26/0/37
qwen3.5-397b	coordinated	19/0/36	28/0/39

Mutation and pairing coverage

The 400 fixture-mutation pairs yielded 0 publications, 300 blocks, and 100 abstentions. Legitimate controls published 40/40; isolated equality-mismatch controls blocked 4/4. Pairing retains 2/4 and 2/4 Gemma families for single/coordinated attacks, and 0/4 and 1/4 Qwen families. Incident-timeline is unpaired in every attack stratum.

The workshop question is what an auditor should record before crediting a system with prevention. We propose retaining the requested action, enforced action, gate reason, and oracle judgment together. This exposes whether prevention came from the supervisor or the boundary, while keeping invalid responses distinct from refusals. The legitimate controls check useful publication as well as blocking; the finite catalog does not establish deployment-wide safety.

Five limits

Scripted wording: 280/280 accepted original worker calls copied the template because `wording_reference` was in the prompt. Unmatched arms: paired contrasts describe the matched families only; zero pairs means unavailable. True summary shown: old provenance observations supplied it when available. Confounded coordination: message counts and speaker roles change together. Four families: the catalog covers only three mechanisms and synthetic local jobs. These observations establish neither learned refusal nor general receipt weakness nor collusion by three speakers.